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Machine Learning Based Operator-Assisted Closed Loop Optimization of Sucker Rod Pumps for Intelligent Automation

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Abstract

Real-time optimization of sucker rod pump operation is essential for maximizing production and reducing operational costs in oilfields. Frequent manual speed adjustments are impractical for fields with thousands of producing wells. This paper addresses these challenges by proposing a semi-closed loop system for intelligent automation of Variable Frequency Drive (VFD) operated sucker rod pumps using machine learning based fillbase prediction. The main goal is to enhance operational efficiency while mitigating risks, by incorporating human oversight for improved safety and reliability in remote oilfield operations. The proposed system integrates VFD-based speed control with ML-driven fillbase prediction (XGBoost), using dynamometer card analysis to determine pump fillage and further adjusts pump speed automatically to achieve the target fillage. A recommendation workflow monitors card quality, speed fluctuations, and alerts operators to anomalies such as persistent bad/no pump cards or significant speed changes. This ensures timely human intervention for safe operations. Various mechanisms like operator-set speed thresholds, snoozing of false recommendations, and feedback capture lead to continuous system refinement. A field trial validates this machine learning based approach and demonstrates its effectiveness in real-world conditions while maintaining operational safety and efficiency. The field trial conducted on 264 wells demonstrated significant improvements across multiple operational metrics. Production analysis revealed that 78 wells (approximately 30% of the test group) achieved an average 35% production uplift through automated speed increases, while 165 wells (62% of the test group) realized 30% power savings from optimized speed reductions. The system's ability to maintain operations around the ideal fillage or "sweet spot" not only enhanced production per unit power consumption, but also improved rod load management, thus reducing mechanical stress and extending equipment lifespan. From an operational standpoint, the solution achieved 90% reduction in manual surveillance requirements, effectively saving one full-time equivalent (FTE) workload while maintaining operational reliability. Notably, the human-in-the-loop feedback mechanism proved particularly useful, with operators intervening in only 12% of the cases where the system flagged potential anomalies or speed constrained operations. These interventions prevented potential issues while allowing the automated system to handle routine optimization. The trial also revealed that the wells operating

at speed limits for extended period benefited from targeted recommendations, enabling operators to make informed adjustments. Overall, the results validate that this semi-automated approach successfully balances the benefits of AI-driven optimization with the critical need for operator oversight in oilfield operations. This hybrid approach enhances production, reduces power consumption, and mitigates operational risks, thus enabling intelligent oilfield operations. By balancing automation with human validation, the system ensures robust operations and high operational efficiency, thus marking a significant advancement in AI-augmented operational excellence in the area of oilfield automation.

Introduction

Sucker Rod Pumps or SRPs are among the most widely used artificial lift systems in the oil and gas industry, particularly in mature and heavy oil fields. These pumps operate by converting rotary motion from a surface motor into vertical reciprocating motion, which lifts reservoir fluids to the surface. Their mechanical simplicity, adaptability to various well conditions, and compatibility with automation technologies make them a preferred choice for long-term production operations (Brown, 1982).

Despite their widespread usage, optimizing sucker rod pump performance remains a complex and labor-intensive task. For optimal production (Huang, 2000), field operators must frequently adjust pump speed to match changing reservoir conditions, regulate fluid levels, or handle gas interference. In large-scale operations with thousands of wells—such as Imperial Oil's Cold Lake operations—manual surveillance and speed tuning are not only time-consuming tasks but are also prone to delays and inconsistencies (Williams, 2000) (Fair, Scott, Boone, Speirs, & Trudell, 2008). This often results in suboptimal production, excessive energy consumption, and increased mechanical wear due to underfill or overfill conditions. For pumping units with Variable Frequency Drives (VFDs) and with proper control algorithms in place, motor speed can be reliably adjusted in automated manner for optimizing production (Chaodong, Pengfei, Xinlun, Haoda, & Ruogu, 2017). However, in such cases too, missing or incorrect pump fillage can lead to inconsistent speed adjustments, thus risking field operations.

To address these challenges, there is a growing need for intelligent automation systems (Kennedy & Ghareeb, 2017) (Martinez, Moreno, Castillo, & Moreno, 1993) that can reliably adjust pump parameters in real-time while maintaining safe operation. The integration of machine learning (ML) algorithms with VFD enabled pumping units offers a promising solution by enabling data-driven decision-making and continuous optimization. ML models can analyze historical and real-time data to predict optimal pump conditions, while VFDs provide the flexibility to implement these adjustments with precision.

Traditional optimization methods rely on rule-based logic, periodic well testing, and manual interpretation of dynamometer cards. While effective in small-scale settings, these approaches lack scalability and responsiveness. Some systems incorporate basic automation through pump-off controllers, which adjust speed based on the surface card analysis. Improved insights into pump operating condition are obtained by translating surface card into downhole card (or pump card) using the wave equation (Gibbs, 2012). However, these systems often require frequent recalibrations and are limited in their ability to adapt to complex or rapidly changing well conditions.

Recent advancements in AI have led to its application in various aspects of oilfield automation, production optimization, anomaly detection, and equipment failure prediction (Kandziora, 2019). In the context of rod pump optimization, machine learning (ML), in particular, has been utilized to analyze pump cards, identify pump related issues and recommend operational adjustments through advanced diagnostics. As shown by few studies (Sharaf, et al., 2019), (Del Pino Fiorillo, 2023), ML based classification of pump cards have enabled their automated interpretation for detecting real-time pump issues like gas interference, fluid pound, leakage etc. An exception-based surveillance system (Al Kiyumi & Al Balushi, 2025) based on ML models can flag potential failures and help in timely action by the field operators. These efforts have demonstrated the potential of ML in improving decision-making accuracy and reducing manual workload.

Figure 1 shows four levels of AI enabled automation from '*NO*' to '*FULL*' automation as we move from left to right. While AI-based automation has been the ongoing trend in the industry, full closed-loop automation (third level) without any operator intervention is risky and could compromise safe operation in case of model errors or data issues. The first level or partial automation is mostly implemented in cases where there are equipment limitations or frequent manual interventions are needed. Human-in-the-loop automation systems (second level) overcome these limitations and achieve higher level of autonomy whereupon operator interventions are needed only as a risk mitigation step.

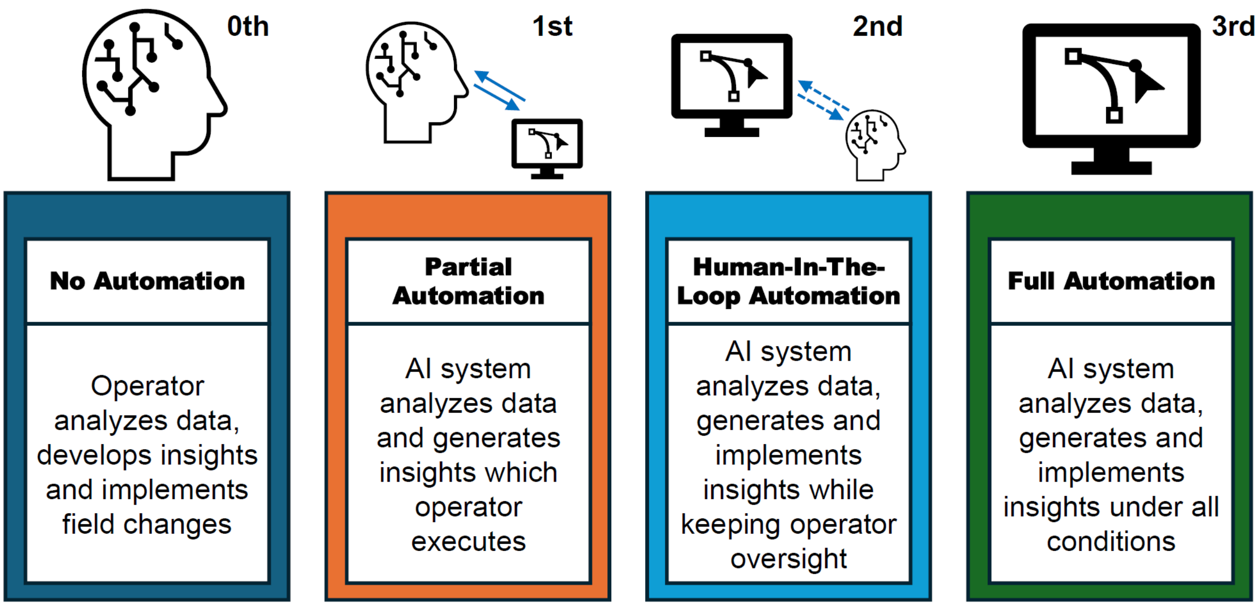

Figure 1—Different levels of AI-led automation ranging from high-to-low needs of operator interventions

As a result, this paper proposes a human-in-the-loop closed-loop system for the real-time optimization of the sucker rod pumps utilized in upstream production operations. As mentioned, VFD enables automated speed adjustments through the machine learning based analysis of the pump cards. In the next section, a brief description of the actual on-field problem is presented along with the challenges related to Cold Lake upstream production. Next, the rod pump optimization strategy is proposed for addressing some of these challenges.

Problem Description

In large-scale oilfield operations like Cold Lake, optimizing sucker rod pump speed is critical for maximizing production efficiency, minimizing power consumption, and reducing mechanical wear. However, manual speed adjustments across thousands of wells are labor-intensive, inefficient, and lack the responsiveness needed to adapt to dynamic reservoir or changing operating conditions. Moreover, in the case of manual adjustments, arriving at the optimal speed for achieving one or several optimization goals is not a trivial task. One of the challenges lies with a highly dynamic relationship between Strokes Per Minute (SPM) and the pump fillage, i.e., the percentage of the pump volume getting filled with the producing fluid. This relationship is difficult to quantify over a wide operational envelope and is required to be learnt by the controller in a dynamic manner to enable autonomous operation of VFDs. Another challenge pertains to the automated determination of the pump fillage, marked by the vertical green line in Figure 2, which requires analysis of the downhole card by the field operation experts. As explained in an earlier paper (Sarma, et al., 2024), this challenge was addressed by developing machine learning based models, trained on historical pump card data, for automated card classification as well as the prediction of the fillbase or load line, shown as horizontal red line in Figure 2. Once the pump fillage and its relationship with pump speed is known, under the autonomous operation, the optimal pump speed is reached to either achieve a target fillage or maximize fluid production.

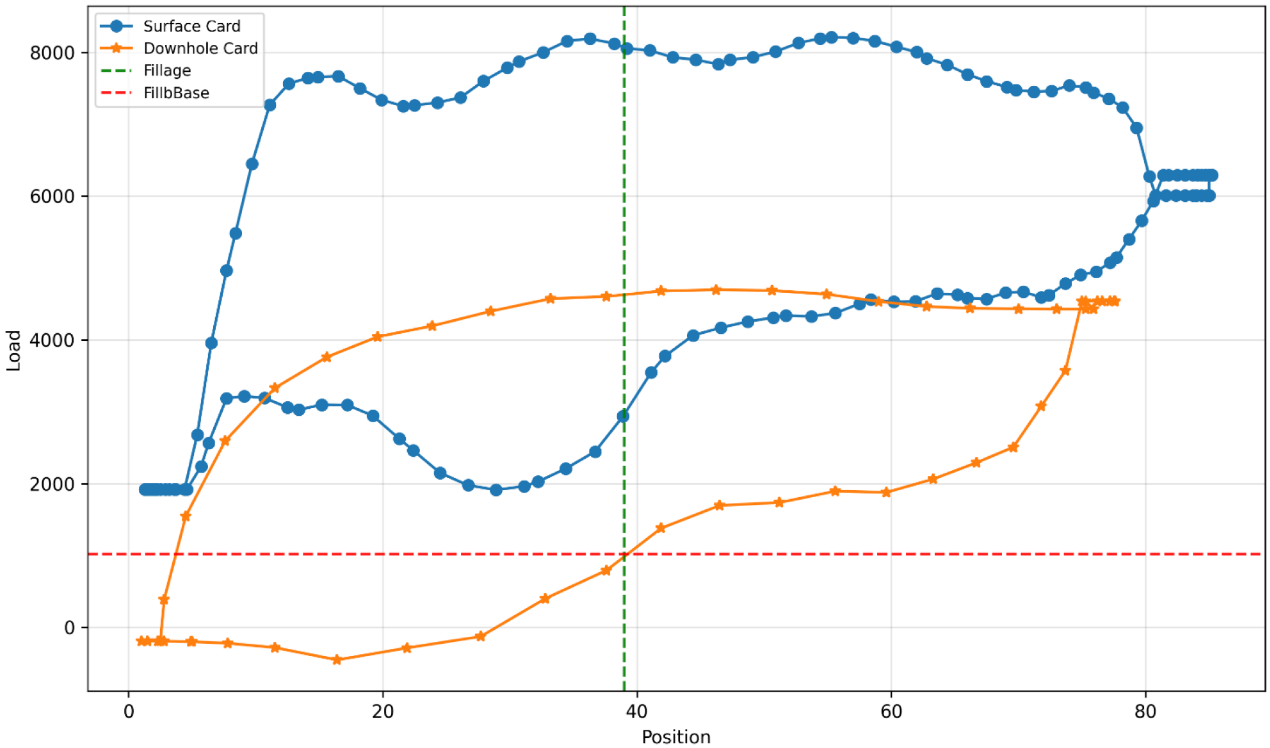

Figure 2—Surface and downhole pump cards with fillage (vertical) and fillbase (horizontal) lines

However, relying solely on a fully automated machine learning driven system introduces additional risks, especially in the edge cases or during unexpected operational changes. This work addresses these operational risks by introducing human assisted closed loop optimization system that leverages ML-based fillbase prediction to automate routine speed adjustments, while incorporating human oversight to review anomalies and intervene as and when necessary. This hybrid approach ensures safe, scalable, and intelligent pump operation by balancing the efficiency of automation with the contextual judgment of experienced operators.

Cold Lake Production Environment

Imperial Oil's Cold Lake upstream operation, located in Alberta, Canada, is one of the world's largest thermal in-situ heavy oil projects. With five processing facilities and over 4,500 producing wells, the site delivers nearly 145,000 barrels of crude oil per day. To overcome high viscosity of bitumen, Cold Lake employs thermal recovery techniques (Vittoratos, Scott, & Beattie, 1990), primarily Cyclic Steam Stimulation (CSS), Figure 3, in the early stages followed by injector-only infills (IOI) and steam flooding once the inter-well communication is established.

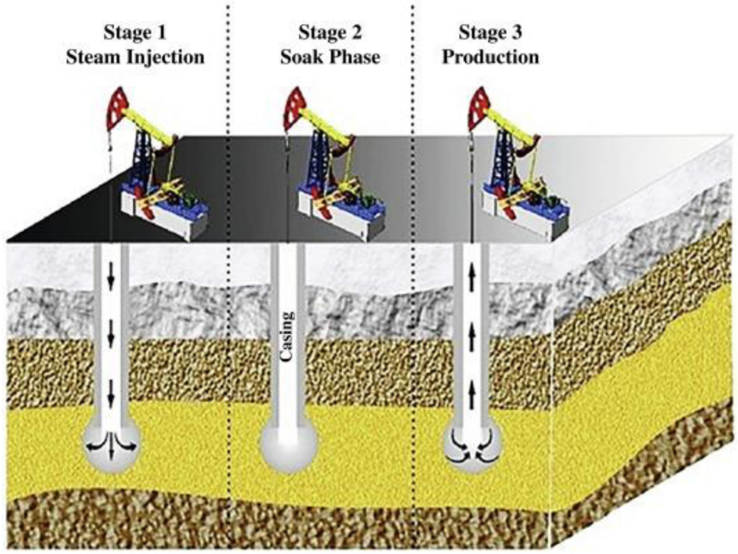

Figure 3—Cyclic Steam Stimulation (CSS) for heavy oil recovery (Hemmati-Sarapardeh, Alamatsaz, Dong, & Li, 2023)

Given the extremely high viscosity of the bitumen and thermal nature of the recovery process, artificial lift is essential for sustained production. Rod pumps are widely used in Cold Lake due to their ability to handle high-temperature fluids, gas interference, and sand production—all are common in thermally stimulated wells. Steam injections raise the temperature of the reservoir to reduce viscosity but also introduce operational challenges that rod pumps are well-suited to manage. Their mechanical simplicity, adaptability to harsh downhole conditions, and compatibility with VFDs make them a reliable choice for artificial lift in this environment.

Among total producing wells in Cold Lake, nearly one-third which produce via rod pumps are VFD enabled, thus providing an opportunity for precise motor speed control for optimizing performance and minimizing energy consumption. The remainder of the wells which require a more conventional method of speed adjustment through manual sheave change is restrictive in nature and allows only a fixed number of discrete speed (SPM) settings. Only VFD enabled rod pumps are under the scope in this work, where autonomous operation can be achieved through the rod pump optimization strategy, as discussed next.

Rod Pump Optimization Strategy

Based on the above discussion, the following five step strategy is proposed for optimizing rod pumps in Cold Lake:

1. Automated classification of pump cards with real-time fillbase prediction[1] using machine learning algorithms.
2. Determination of current pump fillage by the controller using set fillbase line and comparison with the target fillage.
3. Pump speed adjustments within set limits by the Variable Frequency Drive to achieve target fillage or maximum production.
4. System monitoring and recommendation generation in case of anomalous speed changes or other exceptions.
5. Operator intervention with feedback capture for system improvements.

Methodology

Building on the need for scalable and responsive rod pump optimization strategy, this section outlines the design and implementation of a hybrid control system that leverages machine learning for automation while preserving operator oversight. This approach combines data-driven decision-making with field-tested control logic to ensure safe and efficient operation across diverse well conditions.

System Architecture: General Overview of Human Assisted Closed Loop Optimization Framework

A human-assisted closed loop or semi-closed loop optimization system is a hybrid control architecture which blends the speed and scalability of automation with the contextual judgment and safety assurance of human oversight. This structure is particularly suited to complex industrial environments where full automation may not be reliable or safe under all conditions.

At a high level, the system operates in the following stages:

1. **Data Acquisition:** Real-time operational data is collected from field sensors, control systems, or diagnostic tools.
2. **Automated Decision-Making:** A machine learning model or rule-based engine processes data to generate control actions or recommendations.
3. **Control Execution:** These decisions are implemented through actuators or control systems (e.g., adjusting pump speed via VFD).
4. **Monitoring and Exception Handling:** The system continuously monitors performance and flags anomalies, deviations, or edge cases.
5. **Human Oversight:** Operators are notified when the system encounters uncertainty, detects abnormal behavior, or reaches predefined operational boundaries. They can approve, override, or modify the system's actions.
6. **Feedback Loop:** Operator feedback and system performance data are used to refine the model and improve future decision-making.

This architecture, as shown in Figure 4, ensures that the routine operations are handled autonomously, while critical decisions remain under human supervision, hence striking a balance between efficiency, safety, and adaptability.

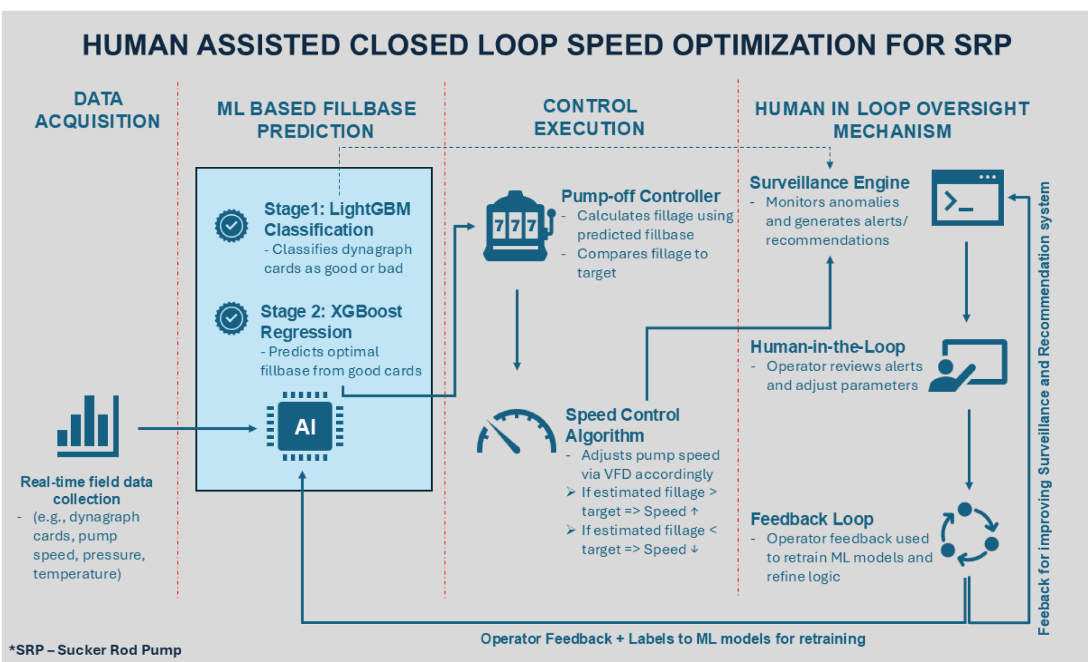

Figure 4—System architecture of the human-assisted closed-loop optimization framework, showing the integration of machine learning-based fillbase prediction, automated speed control via VFDs, and the human-in-the-loop oversight mechanism for anomaly detection, operator intervention and continuous feedback.

Machine Learning-Based Fillbase Prediction

The core of the automation layer in the proposed system is a two-stage machine learning pipeline designed to enhance the accuracy and consistency of fillbase settings—an essential input for pump-off controllers. The first stage is a LightGBM (Ke, et al., 2017) (Friedman & Popescu, 2003) based classification model that evaluates dynamometer cards (position versus load plots) and classifies them as either '*GOOD*' or '*BAD*'. This ensures that only high-quality, reliable cards are used for further analysis. If a card is classified as good, then it is passed on to the second stage: an XGBoost (Chen & Guestrin, 2016) regression model trained to predict the fillbase line or load setting. This prediction is based on a combination of dynagraph features and operational parameters such as pump speed, wellhead pressure, and temperature. The predicted fillbase is then automatically applied to the pump-off controller, which uses it to calculate the pump fillage.

This ML-driven approach replaces manual fillbase tuning, which is time-consuming and inconsistent across wells. Field validation has shown that the classification model achieves ~99% accuracy, while the fillbase prediction model delivers an R-squared of 0.98 and correctly predicts fillbase setting in ~95% of the cases. The result is a robust, scalable solution for real-time optimization across hundreds of wells. The details of the machine learning model, including its architecture, training procedures, and field validation, are comprehensively described in our previous paper (Sarma, et al., 2024), details are skipped here for the sake of brevity.

Speed Control Logic

Once the fillbase is set, the pump-off controller calculates the fillage for each stroke as the new cards get acquired. The system then compares the estimated fillage to a target fillage (typically set by the operator). Based on this comparison, the pump speed is automatically adjusted to maintain optimal operation as explained below:

- If the estimated fillage is below the target, e.g., 50% versus the target of 70%, the pump speed is reduced to allow more time for fluid to enter the pump chamber.
- If the estimated fillage is above the target, the pump speed is increased to improve production efficiency and avoid overfill.
- Under high-inflow conditions, the system prioritizes maximizing fluid lift. In such cases, the pump may operate at higher speeds even if the fillage exceeds the target, ensuring that the production potential is fully captured (Hinojosa, 2023).

This dynamic speed adjustment ensures that the pump operates within its optimal performance window by reducing energy consumption, minimizing mechanical wear and maximizing fluid production.

Human-in-the-Loop Oversight Mechanism

To ensure safety, reliability and adaptability, the system incorporates a human-in-the-loop oversight layer that monitors the automated process and flags exceptions for operator review.

Recommendation Workflow

A surveillance engine continuously evaluates system behavior and generates recommendations or alerts under the following conditions:

- The controller is not producing any dynamometer cards, but the pump is continuously running.
- The system is consistently generating bad cards.

  Example 1. WELL-K1 produced consecutive bad dynagraphs from July 12th to July 14th, shown in Figure 5a). The system flagged this anomaly on July 14th, prompting operator intervention. Following corrective action, the card quality improved on July 15th (Figure 5b), confirming the effectiveness of the recommendation workflow.

- There is a significant deviation in pump speed compared to the previous three days.

Example 2. WELL-K2 maintained good card quality but showed a sudden speed change on July 25th (Figure 6c). The automated system flagged this anomaly on July 26th, prompting operator review. Upon investigation, it was found that the change in speed was driven by a shift in the card shape (Figure 6a), which occurred due to an updated target fillage setting. To achieve the higher fillage, the pump speed was adjusted accordingly, resulting in the fillage estimate shift toward the right on the dynagraph. Since the cards remained consistent on July 26th and 27th (Figure 6b), and the system was operating within expected parameters, no further action was required.

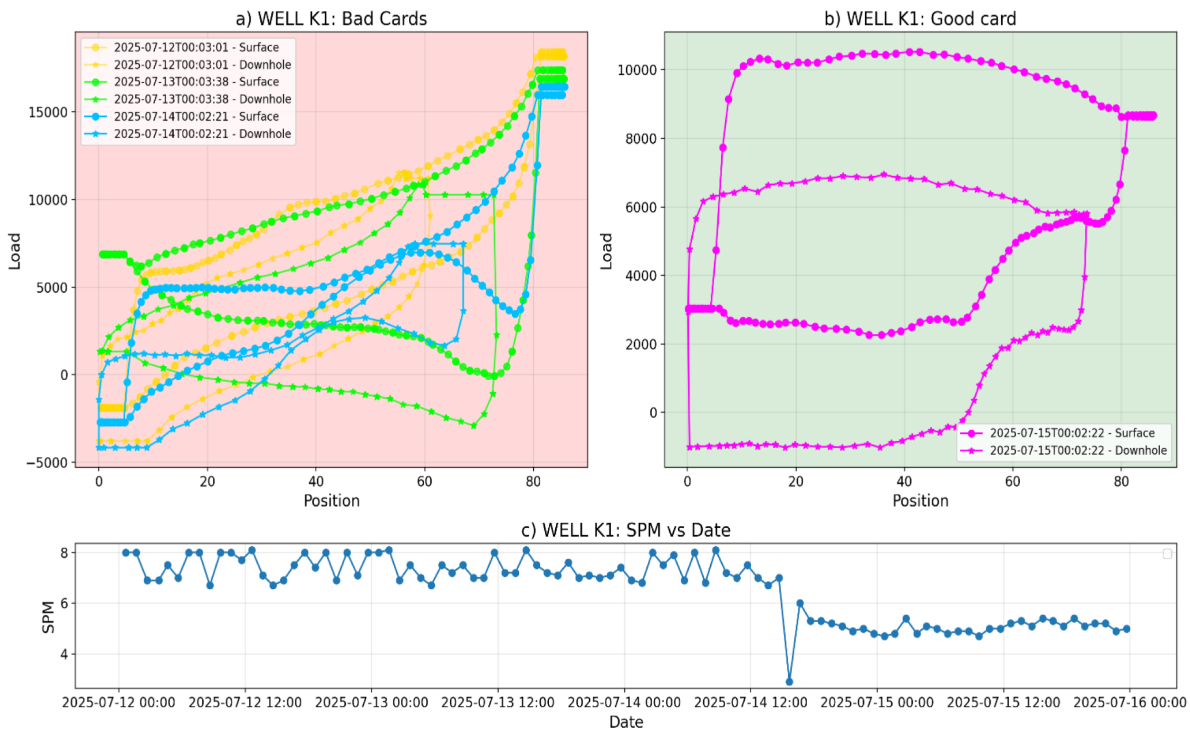

Figure 5—(a) Bad dynagraphs from Well K1 between July 12th–14th (b) Good card observed on July 15th after operator intervention (c) Speed reduction following operator adjustment to improve card shape.

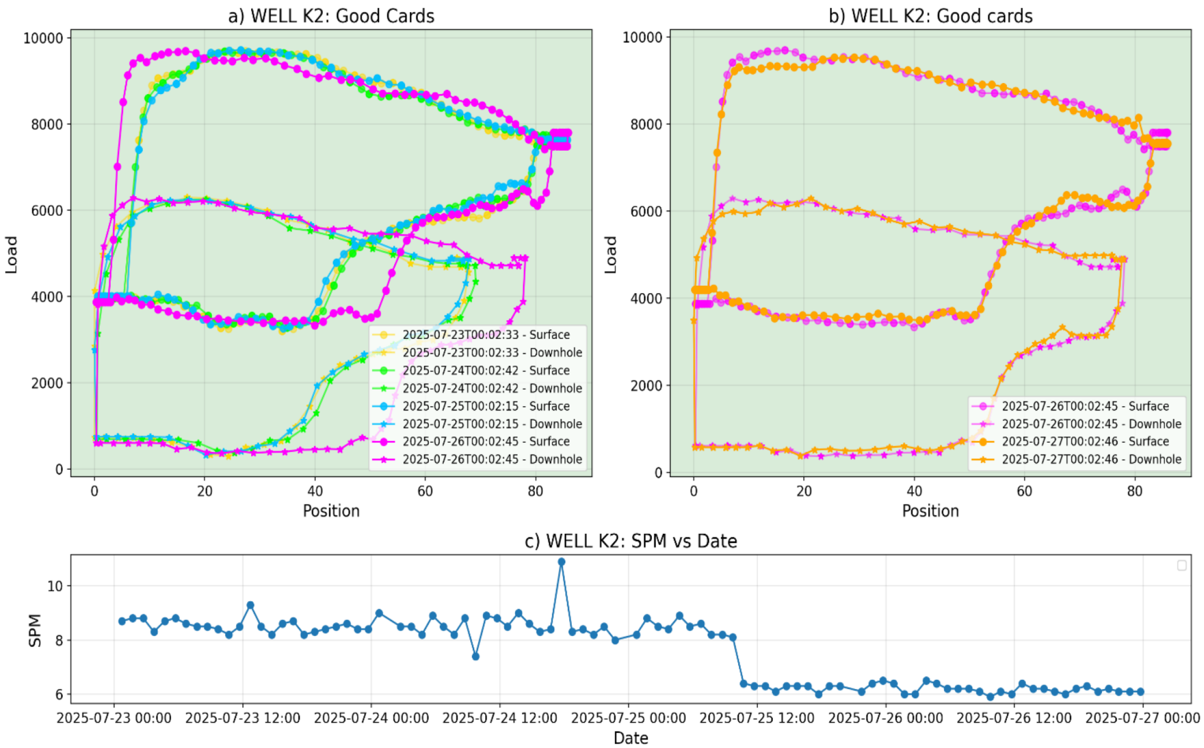

Figure 6—(a) Card shape change observed on July 26th for Well K2. (b) Consistent card quality maintained on July 26th and 27th, confirming no action was needed. (c) Significant speed deviation was detected on July 25th, flagged for operator review.

Operator Intervention Logic

Operators can intervene to perform the following tasks:

- Approve and continue with system-generated speed changes or put the pump at a fixed speed.
- Adjust target fillage or speed thresholds.
- Investigate and resolve issues flagged by the system.

This ensures that while the system handles routine optimization autonomously, operators remain in control of critical decisions and edge cases.

Feedback Loop and System Refinement

Operator feedback post recommendations and corrective actions are captured for the following improvements:

- Refine the machine learning models by providing labeled data for re-training.
- Improve the recommendation logic by identifying patterns in false recommendations or missed anomalies.
- Adapt the system to evolving field conditions and operational strategies.

This feedback loop ensures continuous learning and system improvement, ensuring long-term reliability and performance.

Field Trial And Results

Following the development and deployment of the human-assisted closed-loop optimization system, a comprehensive field trial was conducted to evaluate its on-field performance. The trial focused on 264 producing wells in Cold Lake, all equipped with VFDs. The objective was to assess the impact of the ML-driven, operator-assisted optimization system on production uplift, power savings, and operational efficiency.

The analysis was based on a robust methodology that compared Key Performance Indicators (KPIs) before and after the wells were transitioned to the automated speed control logic. The evaluation assumed that the speed changes were reactive and focused on the relationship between speed/power changes and total fluid production, without accounting for other confounding variables.

Key Performance Metrics

The performance of the closed-loop system was evaluated using two primary Key Performance Indicators (KPIs), each reflecting a critical dimension of operational success:

Production Uplift Without Manual Intervention. In several wells, the system automatically increased pump speed in response to improved inflow conditions—often triggered by steam injection in the neighboring wells. This eventually led to:

- Higher fluid production without requiring manual speed adjustments
- Faster response to transient reservoir conditions
- Reduced risk of underproduction due to delayed operator intervention.

This KPI highlights the system's ability to reactively optimize pump performance in real-time, ultimately ensuring production opportunities are captured promptly and consistently.

Power Efficiency Gains. This category includes both:

- Direct power savings for wells where the system reduced pump speeds during low inflow or decline phases, and
- Improved power-to-production ratio for wells where the energy consumption per unit of fluid produced decreased, even if the total production remained stable. These outcomes demonstrate the system's ability to:
    - Lower operational costs
    - Enhance energy efficiency
    - Sustain production with reduced energy input.

Together, these metrics confirm that the optimization system not only boosts production where possible but also ensures cost-effective and energy-efficient operations across a wide range of well conditions.

Few Examples of Production Uplift and Power Savings

Out of the 264 wells included in the field trial, 78 wells demonstrated a clear production uplift, while 165 wells showed measurable improvements in power efficiency. When analyzed separately, the production uplift group achieved a median increase of approximately 35% in fluid production compared to the pre-automation period. This uplift was primarily driven by the system's ability to respond quickly to improved inflow conditions, often following steam stimulation—by automatically increasing pump speed without requiring manual intervention.

In parallel, the power efficiency group realized a median 30% reduction in energy consumption. These savings were achieved either through direct speed reductions during low-inflow periods or by improving the power-to-production ratio, ensuring more fluid was lifted per unit of energy consumed.

Well-K3 began experiencing increased inflow from May 2025. It was transitioned to automated speed control in July 2024. The system responded by increasing pump speed to lift the additional fluid, resulting in a clear rise in total fluid production without any manual intervention. As shown in Figure 7, this uplift was accompanied by a slight improvement in power efficiency, with power consumption per unit production decreasing compared to the pre-automation period.

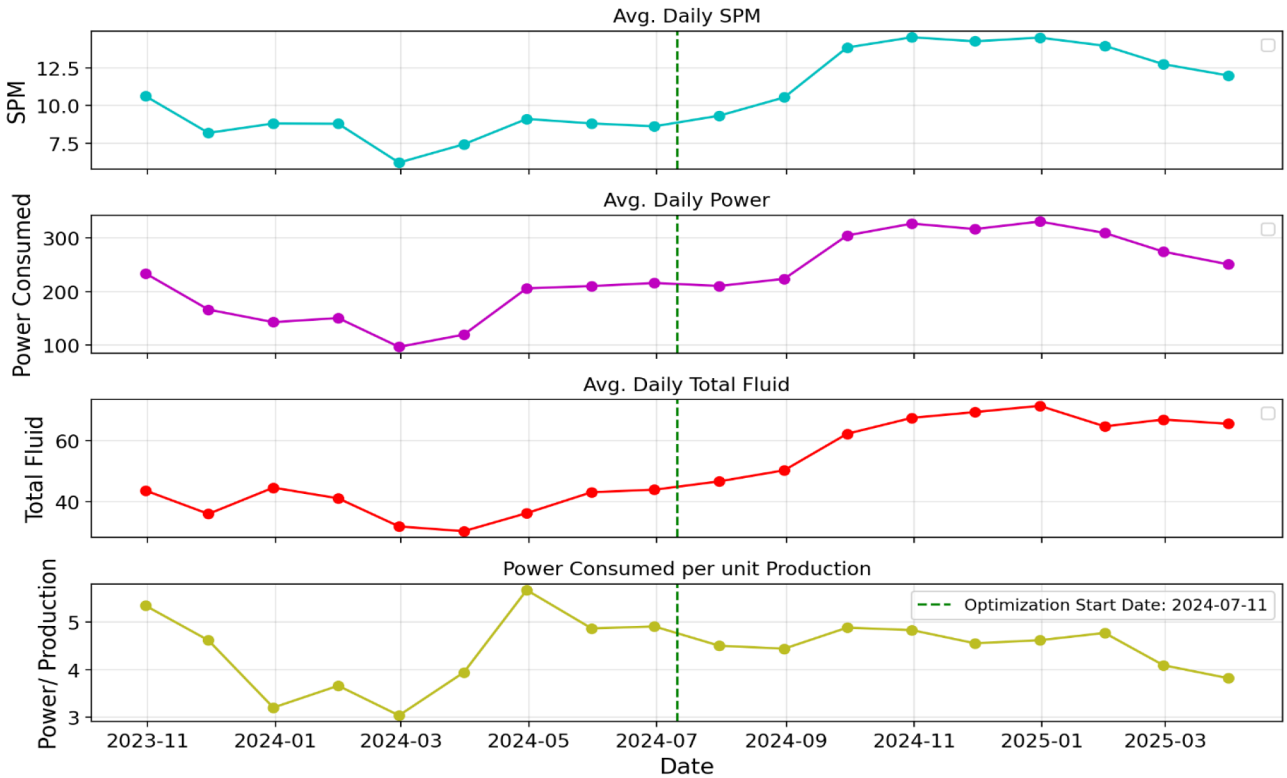

Figure 7—Production uplift case – Well K3. (a) Pump speed (SPM), (b) power consumption, (c) total fluid produced, and (d) power per unit production. The pump was put in automated control in the 1st half of July 2024, leading to increased production and improved efficiency without operator input.

Well-K4 was in a natural production decline phase before automation. Prior to automation, operators manually reduced speed to avoid low fillage. After being placed in automated mode in August 2024, the system continued to reduce speed and power consumption in line with declining inflow. As shown in Figure 8, this resulted in significant power savings, marked by an improvement in power-to-production ratio compared to the pre-automation period.

Together, these results highlight the system's dual capability: to boost production when inflow allows and to conserve energy when conditions are less favorable, delivering tangible value across a wide range of well conditions.

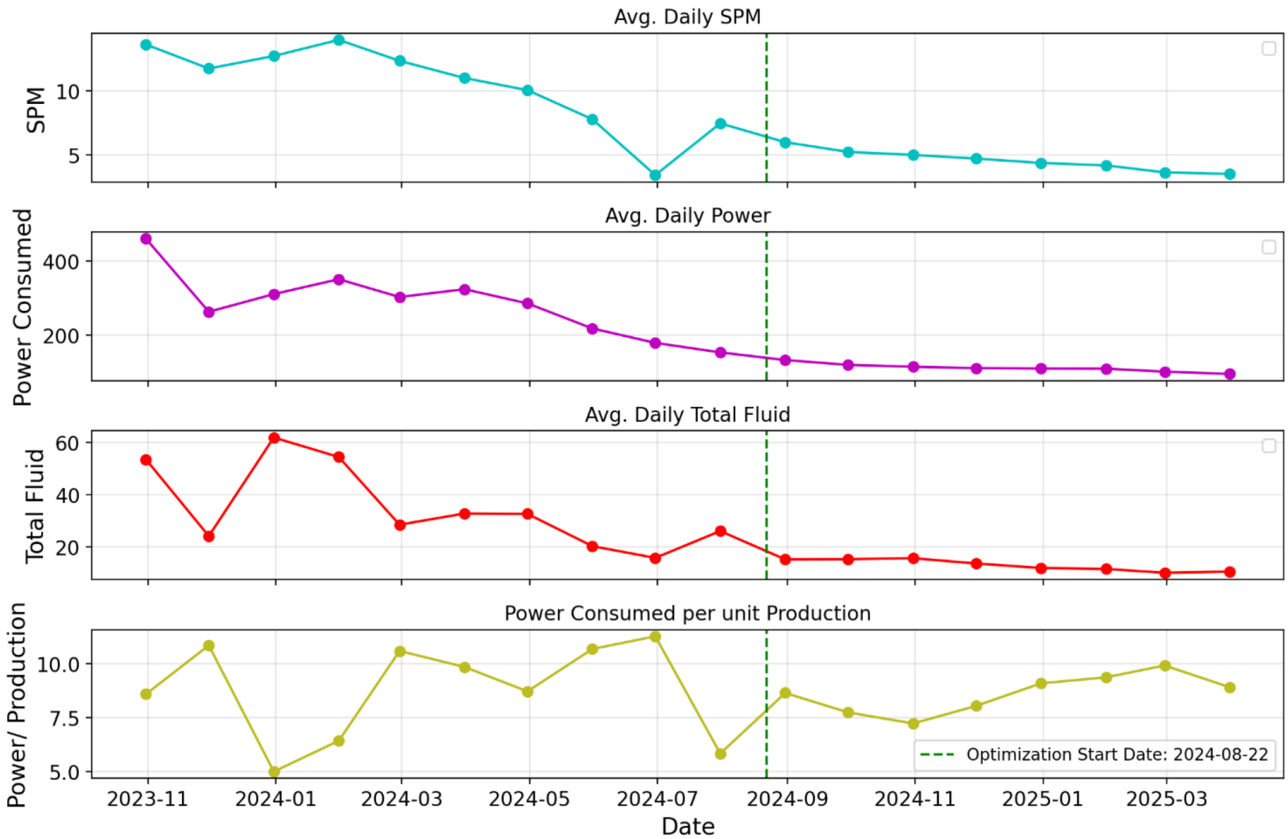

Figure 8—Power savings case – Well K4. ***(a)*** **Pump speed (SPM),** ***(b)*** **power consumption,** ***(c)*** **total fluid produced, and** ***(d)*** **power per unit production. After automation began in August 2024, the system reduced speed and power use, improving energy efficiency during production decline.**

Key Operational Benefits from SRP Intelligent Automation System

Beyond production and power metrics, the optimization system delivered substantial improvements in day-to-day operational efficiency.

- **Reduced Manual Surveillance:** The system's autonomous speed control and anomaly detection capabilities led to a 90% reduction in manual monitoring efforts, effectively saving the workload of one full-time equivalent (FTE). This allowed field personnel to focus on higher-value tasks and exception handling rather than routine speed tuning.
- **Proactive Exception Handling:** The human-in-the-loop mechanism ensured that the operators were engaged only when necessary—such as during persistent bad card generation, missing dynagraphs, or abnormal speed fluctuations. This targeted intervention model improved response times and reduced operational risk.
- **Reduction in Mechanical Wear and Tear of Pumps:** By maintaining pump speed around the ideal fillage or "sweet spot", the system helped in minimizing rod load fluctuations, reducing mechanical wear and extending equipment lifespan.
- **Scalability Across Wells:** The system proved effective across a wide range of well conditions, demonstrating its ability to scale-up without compromising reliability or requiring extensive customization.

Conclusions

This study introduces a scalable closed-loop optimization framework for sucker rod pumps that blends machine learning with operator oversight. Successfully deployed across 264 wells, the system improved production, reduced energy usage, and streamlined operations—demonstrating the value of hybrid automation in complex oilfield environments.

Key outcomes from the field trial are:

- Automated production uplift in the wells responding to reservoir stimulation, eliminating the need for manual speed adjustments.
- Significant power efficiency gains through either direct energy savings or improved power-to-production ratio.
- Substantial reduction in manual surveillance, saving the equivalent of one full-time operator while maintaining operational reliability.

The hybrid architecture proved especially effective in balancing the agility and scalability of intelligent automation by learning from contextual judgment of experienced operators. The system's ability to dynamically adjust pump speeds based on real-time fillage predictions, while escalating anomalies for human review, ensured both safety and adaptability in complex field conditions.

While the results are promising, a few limitations remain. The current system assumes speed changes are reactive and does not explicitly account for all the reservoir or operational variables. Looking ahead, future enhancements could include the integration of predictive optimization models that proactively adjust pump speeds based on anticipated reservoir behavior, rather than reacting solely based on real-time data. The system could also benefit from edge computing capabilities to enable real-time decision-making at the wellsite, reducing latency and dependency on centralized infrastructure. Finally, automating the retraining process of machine learning models using operator feedback and new data streams would ensure continuous learning and adaptability to evolving field conditions.

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